

**ZAGREB PLESO, Croatia**

WMO: 142410

Lat: **45.7292N**Lon: **16.0539E**Elev: **357**StdP: **14.51**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.6       | 17.7 | 7.3                           | 8.1 | 15.8 | 12.4 | 10.5 | 21.1 | 22.4                  | 33.9 | 20.0 | 35.8 | 3.6                   | 250  | 0.459 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.0                   | 91.5            | 71.6 | 88.2 | 70.5 | 85.8 | 69.8 | 73.7                | 86.7 | 72.3 | 84.8 | 70.8 | 82.5 | 6.6                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 69.7                            | 110.8 | 80.1 | 68.1 | 104.5 | 78.6 | 66.5 | 98.7 | 76.5 | 37.5          | 86.8 | 36.1 | 85.1 | 34.9 | 82.7 | 81.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |            |             |                    | n-Year Return Period Values of Extreme Temperature |             |             |             |              |             |              |              |              |     |
|-------------------|-------------|-------------|----------------------------|------------|-------------|--------------------|----------------------------------------------------|-------------|-------------|-------------|--------------|-------------|--------------|--------------|--------------|-----|
| 1%                |             | 2.5%        | 5%                         | Mean       |             | Standard Deviation | n=5 years                                          |             | n=10 years  |             | n=20 years   |             | n=50 years   |              |              |     |
| Min               | Max         | Min         | Max                        | Min        | Max         | Min                | Max                                                | Min         | Max         | Min         | Max          | Min         | Max          | Min          | Max          | DB  |
| (a)               | (b)         | (c)         | (d)                        | (e)        | (f)         | (g)                | (h)                                                | (i)         | (j)         | (k)         | (l)          | (m)         | (n)          | (o)          | (p)          |     |
| (4) <b>19.0</b>   | <b>16.1</b> | <b>13.4</b> |                            | <b>5.2</b> | <b>95.6</b> | <b>6.7</b>         | <b>3.9</b>                                         | <b>0.3</b>  | <b>98.4</b> | <b>-3.6</b> | <b>100.7</b> | <b>-7.4</b> | <b>102.9</b> | <b>-12.2</b> | <b>105.8</b> |     |
| (5)               |             |             | DB                         | <b>4.7</b> | <b>76.6</b> | <b>6.7</b>         | <b>1.9</b>                                         | <b>-0.1</b> | <b>77.9</b> | <b>-4.0</b> | <b>79.0</b>  | <b>-7.7</b> | <b>80.1</b>  | <b>-12.6</b> | <b>81.5</b>  | (p) |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 53.1   | 33.4  | 36.8  | 44.9  | 53.5  | 61.6  | 68.5  | 71.5  | 70.9  | 62.2  | 53.4  | 44.1  | 34.7 |
|                                                                               | DBStd       | 15.23  | 8.20  | 8.57  | 7.82  | 6.57  | 6.09  | 6.07  | 5.22  | 5.46  | 6.03  | 7.37  | 8.03  | 8.07 |
|                                                                               | HDD50       | 1856   | 515   | 372   | 191   | 41    | 2     | 0     | 0     | 0     | 0     | 51    | 207   | 477  |
|                                                                               | HDD65       | 4985   | 979   | 789   | 622   | 346   | 140   | 35    | 9     | 13    | 124   | 362   | 626   | 940  |
|                                                                               | CDD50       | 2972   | 1     | 3     | 34    | 146   | 361   | 555   | 668   | 649   | 366   | 156   | 31    | 2    |
|                                                                               | CDD65       | 625    | 0     | 0     | 0     | 1     | 34    | 140   | 211   | 197   | 39    | 3     | 0     | 0    |
|                                                                               | CDH74       | 6666   | 0     | 0     | 1     | 55    | 444   | 1488  | 2225  | 2027  | 390   | 36    | 0     | 0    |
|                                                                               | CDH80       | 2393   | 0     | 0     | 0     | 5     | 87    | 514   | 885   | 814   | 87    | 1     | 0     | 0    |
| Wind                                                                          | WSAvg       | 4.8    | 4.5   | 5.2   | 6.2   | 5.8   | 5.5   | 4.8   | 4.6   | 4.2   | 4.3   | 4.1   | 4.3   | 4.2  |
| Precipitation                                                                 | PrecAvg     | 34.5   | 2.0   | 1.9   | 2.2   | 2.5   | 3.0   | 3.7   | 3.3   | 3.5   | 3.6   | 3.1   | 3.3   | 2.4  |
|                                                                               | PrecMax     | 48.6   | 6.9   | 5.1   | 5.0   | 6.2   | 7.4   | 7.6   | 8.9   | 10.2  | 9.3   | 8.8   | 10.3  | 5.7  |
|                                                                               | PrecMin     | 20.5   | 0.2   | 0.1   | 0.2   | 0.2   | 0.4   | 1.5   | 1.0   | 0.0   | 0.9   | 0.0   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 6.2    | 1.3   | 1.2   | 1.0   | 1.3   | 1.5   | 1.3   | 1.7   | 2.2   | 1.9   | 2.1   | 2.0   | 1.3  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 58.7  | 62.9  | 71.9  | 79.9  | 86.2  | 92.8  | 96.0  | 96.9  | 87.5  | 78.4  | 69.6  | 59.4 |
|                                                                               |             | MCWB   | 51.3  | 50.6  | 54.7  | 61.7  | 67.7  | 72.6  | 71.1  | 69.7  | 67.3  | 64.2  | 58.9  | 52.1 |
|                                                                               | 2%          | DB     | 53.7  | 58.6  | 66.7  | 75.0  | 82.4  | 89.2  | 91.4  | 91.6  | 82.3  | 73.7  | 64.1  | 54.2 |
|                                                                               |             | MCWB   | 47.8  | 49.0  | 52.7  | 58.6  | 65.7  | 71.7  | 72.0  | 71.3  | 67.5  | 63.1  | 56.3  | 49.1 |
|                                                                               | 5%          | DB     | 49.7  | 54.0  | 62.8  | 71.4  | 79.0  | 85.9  | 88.0  | 88.0  | 78.7  | 69.7  | 59.4  | 50.1 |
|                                                                               |             | MCWB   | 46.0  | 46.7  | 51.4  | 57.4  | 64.1  | 70.2  | 70.8  | 70.0  | 65.7  | 61.4  | 54.0  | 46.8 |
|                                                                               | 10%         | DB     | 45.0  | 50.1  | 58.9  | 67.6  | 75.4  | 82.5  | 85.2  | 84.4  | 74.9  | 65.6  | 55.7  | 46.0 |
|                                                                               |             | MCWB   | 42.3  | 44.4  | 49.5  | 55.2  | 62.3  | 68.4  | 69.8  | 68.9  | 64.1  | 59.0  | 52.2  | 44.2 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.8  | 52.2  | 56.1  | 63.4  | 69.9  | 75.5  | 75.2  | 75.6  | 70.5  | 66.4  | 59.4  | 53.0 |
|                                                                               |             | MCDB   | 57.0  | 60.8  | 67.9  | 76.2  | 81.9  | 88.0  | 88.0  | 88.3  | 81.3  | 75.2  | 66.4  | 57.8 |
|                                                                               | 2%          | WB     | 48.7  | 49.7  | 54.0  | 60.1  | 67.4  | 73.2  | 73.6  | 73.3  | 68.4  | 63.9  | 57.0  | 49.9 |
|                                                                               |             | MCDB   | 53.1  | 57.4  | 64.6  | 71.7  | 79.5  | 85.7  | 86.8  | 86.3  | 79.3  | 71.7  | 62.5  | 53.8 |
|                                                                               | 5%          | WB     | 45.9  | 47.4  | 52.0  | 58.0  | 65.2  | 71.2  | 72.3  | 71.6  | 66.6  | 61.8  | 54.9  | 46.8 |
|                                                                               |             | MCDB   | 49.1  | 53.4  | 61.1  | 68.6  | 76.6  | 83.2  | 84.9  | 83.7  | 76.7  | 68.3  | 59.4  | 49.7 |
|                                                                               | 10%         | WB     | 42.5  | 44.9  | 50.1  | 56.1  | 63.2  | 69.3  | 70.7  | 70.1  | 64.9  | 59.6  | 52.2  | 43.8 |
|                                                                               |             | MCDB   | 44.6  | 49.3  | 57.8  | 65.3  | 73.7  | 80.5  | 82.4  | 81.7  | 73.3  | 64.9  | 56.2  | 45.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 12.5  | 16.6  | 19.6  | 20.9  | 21.3  | 21.2  | 22.0  | 22.4  | 19.9  | 18.1  | 13.8  | 11.8 |
|                                                                               |             | MCDBR  | 19.7  | 24.6  | 27.5  | 29.1  | 27.6  | 26.4  | 27.3  | 28.7  | 26.0  | 23.9  | 21.0  | 18.8 |
|                                                                               |             | MCWBR  | 14.4  | 15.9  | 15.4  | 15.1  | 13.2  | 11.6  | 10.7  | 10.9  | 12.2  | 13.9  | 14.1  | 14.2 |
|                                                                               | 5% WB       | MCDBR  | 18.3  | 22.3  | 23.9  | 25.3  | 25.1  | 24.9  | 24.6  | 25.1  | 23.8  | 21.8  | 19.4  | 17.7 |
|                                                                               |             | MCWBR  | 13.7  | 14.9  | 13.7  | 13.7  | 12.7  | 11.4  | 10.8  | 11.0  | 11.9  | 13.3  | 13.3  | 13.6 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.337  | 0.363 | 0.399 | 0.448 | 0.436 | 0.440 | 0.439 | 0.440 | 0.426 | 0.398 | 0.364 | 0.336 |      |
|                                                                               | taud        | 2.347  | 2.287 | 2.228 | 2.111 | 2.195 | 2.229 | 2.229 | 2.239 | 2.246 | 2.307 | 2.351 | 2.383 |      |
|                                                                               | Ebn at Noon | 238    | 254   | 260   | 257   | 265   | 264   | 262   | 257   | 249   | 238   | 225   | 225   |      |
|                                                                               | Edh at Noon | 25     | 31    | 38    | 47    | 45    | 43    | 43    | 41    | 37    | 31    | 24    | 22    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 390    | 629   | 1009  | 1385  | 1717  | 1933  | 1945  | 1711  | 1180  | 771   | 428   | 318   |      |
|                                                                               | RadStd      | 60     | 107   | 139   | 159   | 158   | 140   | 127   | 186   | 153   | 89    | 58    | 50    |      |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | +0.94 | N/A     | N/A    | +1.85   | N/A   | N/A   | N/A         | N/A   | +175  | +112  |
| Regional (50 neighbors) | +0.83 | N/A     | N/A    | +1.48   | +0.81 | +0.68 | N/A         | -211  | +157  | +101  |